Sequences and Series

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Sequence limits**

Determine whether each sequence converges, and find the limit if it does:

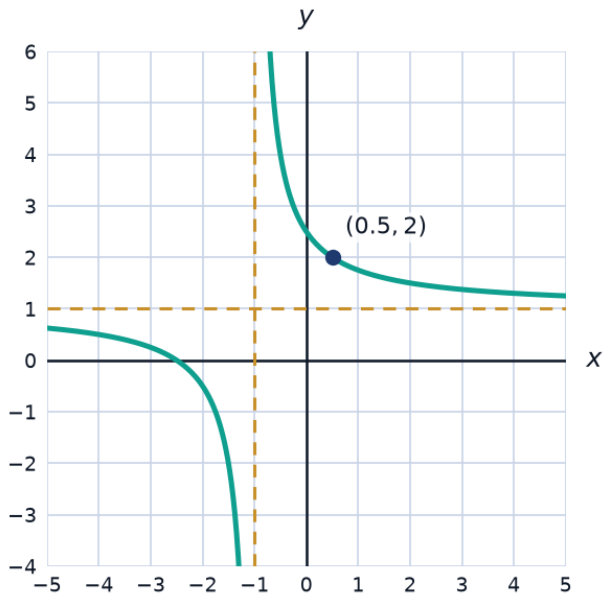
a $a_n = \frac{3n+1}{n+2}$

b $a_n = \frac{n^2}{2^n}$

c $a_n = (-1)^n$

2 **[Foundational]** • **Target: Sequence limits**

Evaluate $\lim_{n\to\infty} \left(1+\frac{3}{n}\right)^n$.



3 **[Foundational]** • **Target: Series and partial sums**

Determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges or diverges, and name the test or fact that justifies your answer.

4 **[Foundational]** • **Target: Series and partial sums**

Evaluate the geometric series $\sum_{n=0}^{\infty} 3\left(\frac{2}{5}\right)^n$.

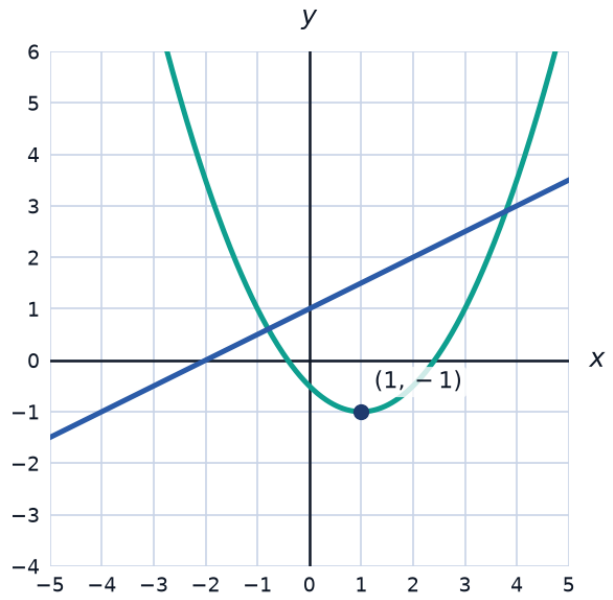
5 **[Foundational]** • **Target: Series and partial sums**

Use partial fractions and telescoping to evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

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6 [Foundational] • Target: Monotonic and bounded sequences

Show that the sequence $a_n = \frac{n}{n+1}$ is increasing, by comparing a_{n+1} and a_n.



7 [Foundational] • Target: Monotonic and bounded sequences

Explain why the sequence $a_n = \frac{n}{n+1}$ (increasing and bounded above by 1) must converge, and find its limit.

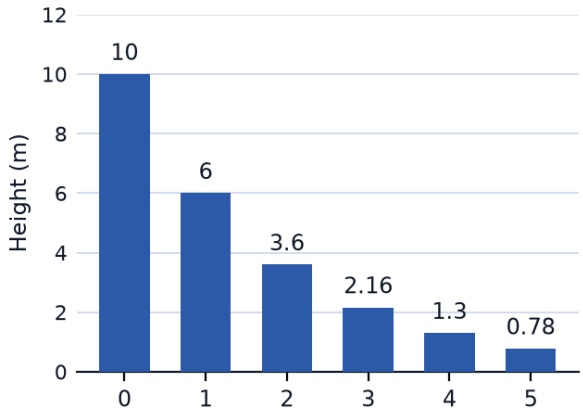
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More series practice

Evaluate the series $\sum_{n=1}^{\infty} \left[\left(\frac{1}{2}\right)^n + \left(\frac{1}{3}\right)^n\right]$ by splitting into two geometric series.

9 [Intermediate] • Target: More series practice

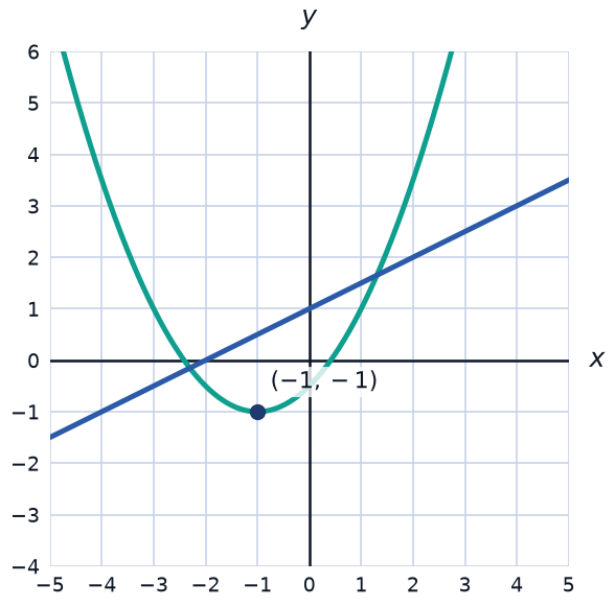
A ball dropped from 10 m rebounds to 60% of its previous height each bounce. Using an infinite geometric series, find the total distance traveled.



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10 [Intermediate] • Target: Recursive sequences

A sequence is defined recursively by $a_1 = 2$, $a_{n+1} = \frac{a_n + 6}{2}$. Compute a_2, a_3, and a_4.



11 [Intermediate] • Target: Recursive sequences

Assuming the recursive sequence above converges to a limit L, find L by setting $L = \frac{L + 6}{2}$ and solving.

12 [Intermediate] • Target: Repeating decimals as series

Express the repeating decimal $0.\overline{45}$ as a fraction using an infinite geometric series with $a = \frac{45}{100}$ and $r = \frac{1}{100}$.

13 [Intermediate] • Target: Visualizing partial sums of a series

The graph shows the first six partial sums $S_n = \sum_{k=1}^{n} \left(\frac{1}{2}\right)^k$ of a geometric series. Use the pattern to conjecture the value the sums are approaching, and confirm it using the geometric series formula.

14 [Intermediate] • Target: Series with factorial and combined terms

Evaluate $\sum_{n=0}^{\infty} \frac{1}{2^n} - \sum_{n=0}^{\infty} \frac{1}{3^n}$ by evaluating each geometric series separately.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

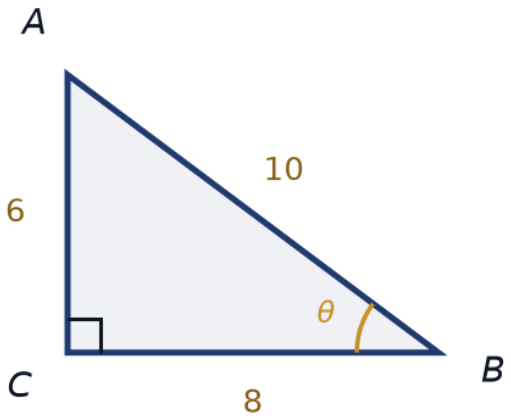
15 [Challenging] • Target: Series with factorial and combined terms • Cross-Lesson: 6.02 Optimization and Curve Sketching

A sequence is defined by $a_n = \frac{2n + (-1)^n}{n}$. Find $\lim_{n \to \infty} a_n$, explaining why the $(-1)^n$ term does not affect the limit.

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16 [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 7.01 Integration by Substitution

Use the squeeze theorem to find $\lim_{n\to\infty} \frac{\sin n}{n}$, using the fact that $-1 \le \sin n \le 1$ for all n.



17 [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 7.02 Integration by Parts

Use the squeeze theorem to find $\lim_{n\to\infty} \frac{n!}{n^n}$, using the bound $0 < \frac{n!}{n^n} \le \frac{1}{n}$ for $n \ge 1$.

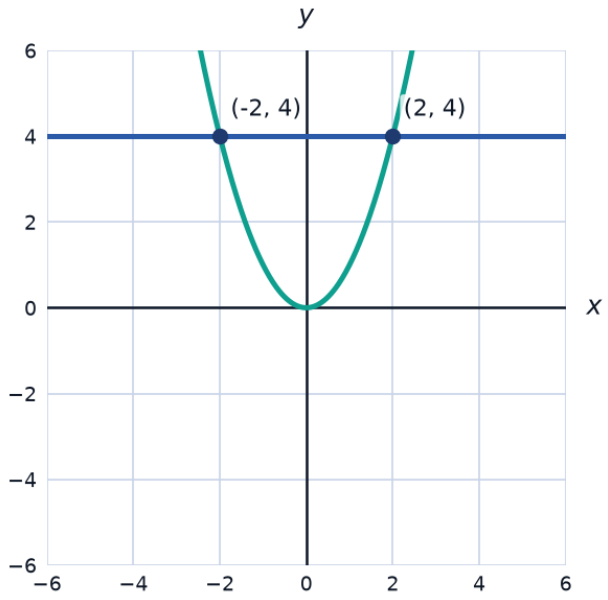
18 [Challenging] • Target: Applications of infinite series • Cross-Lesson: 8.01 Area Between Curves

A patient takes a 100 mg dose of medication every 8 hours. Between doses, 75% of the drug remaining is eliminated, so a fraction $r = 0.25$ carries over to just before the next dose. Using an infinite geometric series, find the long-run steady-state amount in the body immediately after a dose (assume doses accumulate indefinitely).

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19 [Challenging] • Target: Recursive sequences • Cross-Lesson: 8.02 Volumes of Revolution

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.



20 [Challenging] • Target: Repeating decimals as series • Cross-Lesson: 8.03 Arc Length and Improper Integrals

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ needed to guarantee an error less than $\frac{1}{1000}$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Limits of Sequences

Find the limit of the sequence $a_n = \frac{3n^2 - 5n}{2n^2 + 7}$ as $n \to \infty$.

a. 3/2

b. 0

c. ∞

d. 3/7

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22 [Foundational] • Target: Limits of Sequences

Determine if the sequence $a_n = \frac{(-1)^n}{n}$ converges, and if so, find its limit.

a. Converges to 0

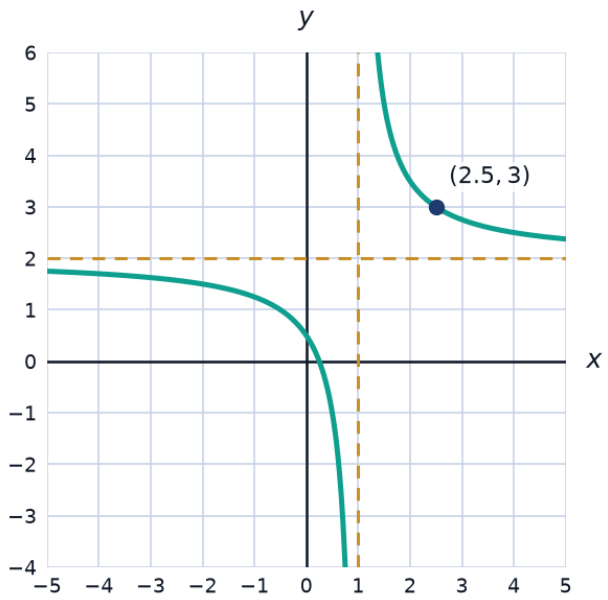
b. Diverges (oscillates)

c. Converges to 1

d. Converges to -1

23 [Foundational] • Target: Limits of Sequences

Find the limit of the sequence $a_n = \left(1 + \frac{2}{n}\right)^n$ as $n \to \infty$.



a. e^2

b. e

c. 1

d. ∞

24 [Foundational] • Target: Geometric Series

Find the sum of the geometric series $\sum_{n=1}^{\infty} 3\left(\frac{2}{5}\right)^{n-1}$.

a. 5

b. $15/2$

c. 3

d. 15

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25 [Foundational] • Target: Geometric Series

Determine the convergence of $\sum_{n=0}^{\infty} 4\left(-\frac{3}{2}\right)^n$.

a. Diverges because $|r| = 3/2 \geq 1$

b. Converges to $-8/5$

c. Converges to $8/5$

d. Converges to 0

26 [Foundational] • Target: Geometric Series

Express the repeating decimal $0.\overline{27} = 0.272727\ldots$ as a simplified fraction using a geometric series.

a. $3/11$

b. $27/100$

c. $9/33$

d. $27/99$ is $3/11$

27 [Foundational] • Target: Telescoping Series

Find the sum of the telescoping series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

a. 1

b. $1/2$

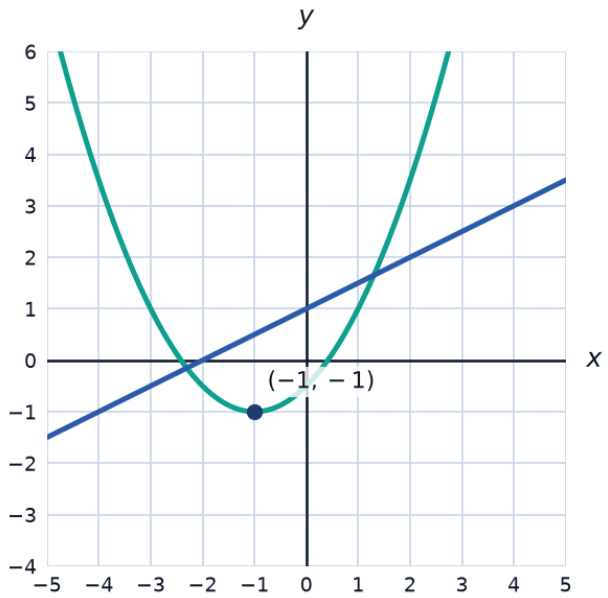
c. 2

d. Diverges

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28 [Foundational] • Target: Telescoping Series

Find the sum of $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right)$.



a. 1

b. 0

c. ∞

d. $\sqrt{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Telescoping Series

Evaluate $\sum_{n=1}^{\infty} \ln\left(\frac{n+1}{n}\right)$.

a. Diverges to ∞

b. 0

c. 1

d. $\ln 2$

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30 [Intermediate] • Target: Divergence Test (nth-Term Test)

What does the nth-Term Test for Divergence state?

a. If $\lim_{n\to\infty} a_n \neq 0$, then $\sum a_n$ diverges

b. If $\lim_{n\to\infty} a_n = 0$, then $\sum a_n$ converges

c. If $\sum a_n$ diverges, then $\lim_{n\to\infty} a_n \neq 0$

d. If $\lim_{n\to\infty} a_n = 0$, the test proves divergence

31 [Intermediate] • Target: Divergence Test (nth-Term Test)

Apply the Divergence Test to $\sum_{n=1}^{\infty} \frac{n}{2n+1}$.

a. Diverges because $\lim_{n\to\infty} a_n = 1/2 \neq 0$

b. Converges to $1/2$

c. Inconclusive

d. Converges by p-series

32 [Intermediate] • Target: Divergence Test (nth-Term Test)

Why is the Divergence Test inconclusive for the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$?

a. Because $\lim_{n\to\infty} \frac{1}{n} = 0$, which does not guarantee convergence

b. Because $1/n$ is negative

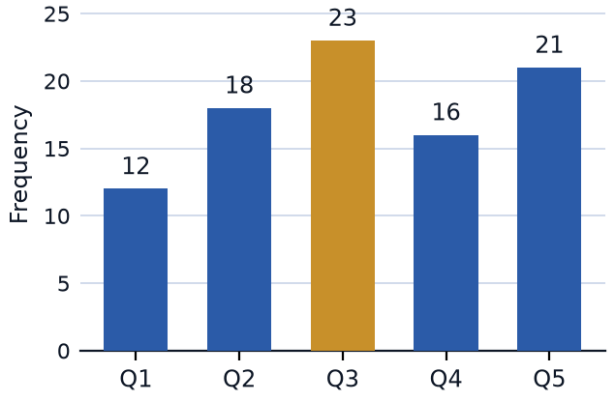
c. Because the terms oscillate

d. Because it is a geometric series

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33 [Intermediate] • Target: Telescoping Series

The sequence of partial sums $S_n = 2 - 1/2^n$: this means $\sum a_n =$



a. 1

b. 3/2

c. 2

d. diverges

34 [Intermediate] • Target: Divergence Test (nth-Term Test)

Dirichlet test: $\sum a_n b_n$ converges if a_n monotone $\to 0$ and partial sums of $\sum b_n$ are:

a. convergent

b. zero

c. increasing

d. bounded

35 [Intermediate] • Target: Sequence limits

Consider the arithmetic sequence with the first term $a_1 = 5$ and a common difference $d = 3$. What is the value of the 5th term, a_5?

a. 17

b. 20

c. 15

d. 23

36 [Intermediate] • Target: Series and partial sums

A monotone bounded sequence:

a. diverges

b. converges

c. oscillates

d. is arithmetic

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37 [Intermediate] • Target: Monotonic and bounded sequences

An increasing sequence bounded above:

a. diverges

b. oscillates

c. converges

d. is arithmetic

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: More series practice • Cross-Lesson: 8.01 Area Between Curves

Sum of infinite geometric series ($|r| < 1$): $\sum_{k=0}^{\infty} ar^k =$

a. $\frac{a}{r}$

b. $\frac{a}{1-r}$

c. $a(1-r)$

d. $\frac{a}{1+r}$

39 [Challenging] • Target: Recursive sequences • Cross-Lesson: 8.02 Volumes of Revolution

Sum of n terms of geometric $1 + 2 + 4 + \cdots$: $S_n =$

a. $2^n - 1$

b. 2^n

c. $n \cdot 2^n$

d. 2^{n+1}

40 [Challenging] • Target: Repeating decimals as series • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Geometric series $3 + 1 + 1/3 + 1/9 + \cdots$: first term $a = 3$, $r =$

a. 1/3

b. 1/2

c. 3

d. 1/9

41 [Challenging] • Target: Visualizing partial sums of a series • Cross-Lesson: 9.01 Separable Differential Equations

Geometric sequence: $a_n = a_1 \cdot r^{n-1}$. If $a_1 = 2, r = 3$, then $a_4 =$

a. 6

b. 18

c. 54

d. 162

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42 [Challenging] • Target: Series with factorial and combined terms • Cross-Lesson: 1.01 Precalculus Review for Calculus

Squeeze theorem for sequences: if $a_n \le b_n \le c_n$ and $a_n, c_n \to L$, then $b_n \to$

a. L

b. $L/2$

c. 0

d. $2L$

43 [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

A sequence is bounded above if there exists M such that $a_n \le M$ for all n. This bound is:

a. always 0

b. the limit

c. not necessarily unique

d. equal to the supremum

44 [Challenging] • Target: Applications of infinite series • Cross-Lesson: 12.01 Convergence Tests

Geometric sequence with $a_1 = 5$, ratio $r = -1/2$: does it converge?

a. No

b. Yes, to -5

c. Yes, to 5

d. Yes, to 0

45 [Challenging] • Target: Limits of Sequences • Cross-Lesson: 12.02 Power Series and Taylor Series

The n-th term test: if $\lim_{n\to\infty} a_n e 0$, the series $\sum a_n$:

a. converges

b. oscillates

c. may converge

d. diverges

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